1. Why CVP fails

Limitations

- Static pressure ≠ volume; poor predictor of fluid responsiveness (Marik 2013: AUC ≈ 0.56)
- Affected by RV function, tricuspid disease, PEEP, intra-abdominal pressure, compliance, tamponade
- High CVP (e.g. 16) with oliguria → venous congestion lowers renal perfusion pressure (MAP − CVP)

Where CVP helps

- Trend, RV failure, tamponade, safety ceiling for fluids, ScvO2 sampling, waveform (cannon a, large v)

Key point: "CVP tells you about the right heart and congestion, not about fluid need."

2. Fluid responsiveness

Dynamic tests

- **Passive leg raise**: ↑ CO/VTI ≥ 10% (works in AF, spontaneous breathing)
- PPV/SVV > 12–13% only if ventilated Vt ≥ 8, sinus rhythm, closed chest, no spontaneous effort
- End-expiratory occlusion (↑ CO ≥ 5%), mini-fluid challenge 100 mL, tidal volume challenge
- IVC variability (limited; spontaneous breathing unreliable)

Key point: Responsive ≠ needs fluid: give only if responsive AND hypoperfused.

3. Advanced monitoring compared

Pulse contour (FloTrac, LiDCO, PiCCO)

- Continuous CO and SVV; uncalibrated less accurate in vasoplegia, arrhythmia, IABP, poor arterial trace

Transpulmonary thermodilution (PiCCO, VolumeView)

- Calibrated CO, GEDV (preload), EVLW (oedema), PVPI; needs central line + femoral arterial; recalibrate

Pulmonary artery catheter

- CO, PAOP, PAP, SvO2, PVR; RV failure, pulmonary hypertension, cardiac surgery
- PAC-Man 2005: no mortality benefit; complications

Echocardiography

- Diagnosis: LV/RV function, tamponade, valves, LVOT obstruction (SAM), VTI
- Operator-dependent, intermittent; TOE after cardiac surgery

Key point: Echo to diagnose, a continuous CO monitor to titrate.

4. Integrating the data: shock phenotype

Patterns

- **LV failure**: ↓ CO/CI < 2.2, ↑ PAOP > 18, ↑ SVR, ↓ SvO2; echo poor LVEF
- **RV failure**: ↑ CVP > PAOP, dilated RV, D-shaped septum, TAPSE < 17 mm, ↑ PAP
- **Tamponade**: equalised pressures, collapsing RA/RV, low CO; localised clot after surgery
- **Vasoplegia**: normal/↑ CO, ↓ SVR
- **Hypovolaemia**: ↓ CO, small hyperdynamic LV, responsive to PLR

Key point: Name the phenotype, then match the drug to the problem.

5. Treatment by phenotype

LV failure

- Optimise preload, inotrope (dobutamine/milrinone), noradrenaline for MAP, reduce afterload when MAP allows
- Mechanical support: IABP (IABP-SHOCK II: no benefit in MI), Impella, VA-ECMO (ECLS-SHOCK: no benefit routinely)

RV failure

- Avoid overload; maintain MAP > PAP (noradrenaline, vasopressin)
- Milrinone/dobutamine, inhaled NO/epoprostenol, lower PEEP/plateau, correct hypoxia, hypercapnia, acidosis

Tamponade

- Re-sternotomy (cardiac surgery) or pericardiocentesis; keep preload and HR, avoid high PEEP

Key point: RV failure: "pressure up, pulmonary resistance down, avoid fluids".

6. Post-cardiotomy shock checklist

Causes

- Bleeding/hypovolaemia, tamponade, LV/RV dysfunction (stunning, graft failure), valve dysfunction/SAM after MV repair, vasoplegia, arrhythmia, pacing problems

Actions

- Check drains, ROTEM, echo (TOE), ECG, pacing wires (AV pacing 90)
- Methylene blue for refractory vasoplegia; CALS for arrest (resternotomy < 5 min)

Key point: After cardiac surgery: think bleeding and tamponade first.

7. Cardiogenic shock (SCAI)

SCAI stages

- A at risk, B beginning, C classic, D deteriorating, E extremis

Key steps

- Early revascularisation of culprit lesion (SHOCK trial); culprit-only PCI (CULPRIT-SHOCK)
- Lactate and end-organ monitoring; shock team; noradrenaline > dopamine (SOAP II)

Key point: Revascularise early, support wisely, avoid routine mechanical devices.